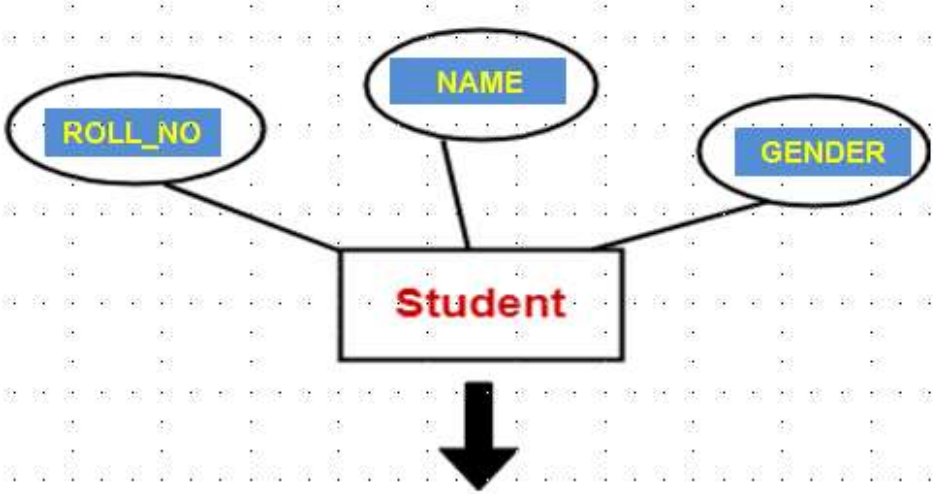
Converting ER-Diagrams to Tables

- ***Relational Model:*** ***ER diagram is converted into the tables after design***
- ***Easily implemented by RDBMS like MySQL , Oracle etc***
- ***Follows certain rules for making this conversion possible***

Explanation with Example



A **strong entity set with only simple attributes** will require **only 1 table** in relational model.

Attributes of the table will be the attributes of the entity set.

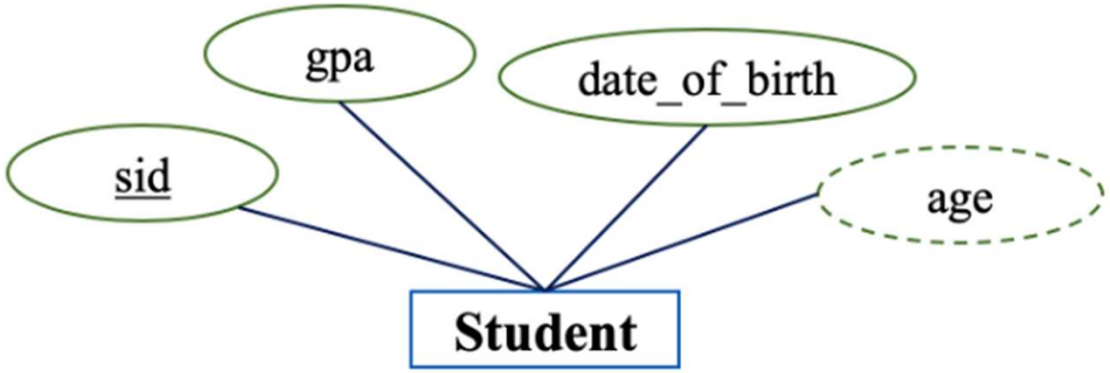
Derived attributes removed from the table.

The primary key of the table will be the key attribute of the entity set.

ROLL_NO	NAME	GENDER

Schema : Student (ROLL_NO , NAME , GENDER)

Explanation with Example



sid	gpa	date_of_birth

Schema: Student(sid, gpa, date_of_birth)

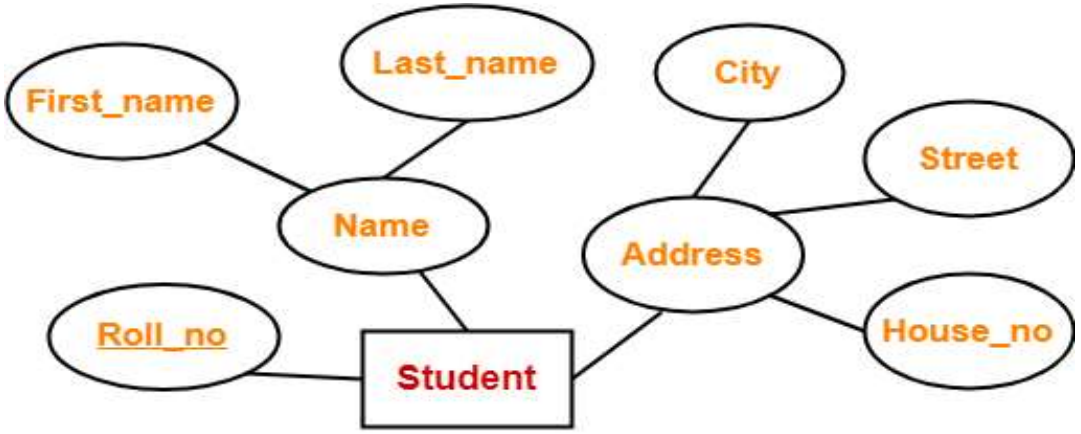
Rules for Conversion

Rule-02: For Strong Entity Set With Composite Attributes-

➢ *A strong entity set with any number of composite attributes will require only one table in relational model.*

➢ *While conversion, simple attributes of the composite attributes are taken into account and not the composite attribute itself.*

Explanation with Example



Roll_no	First_name	Last_name	House_no	Street	City

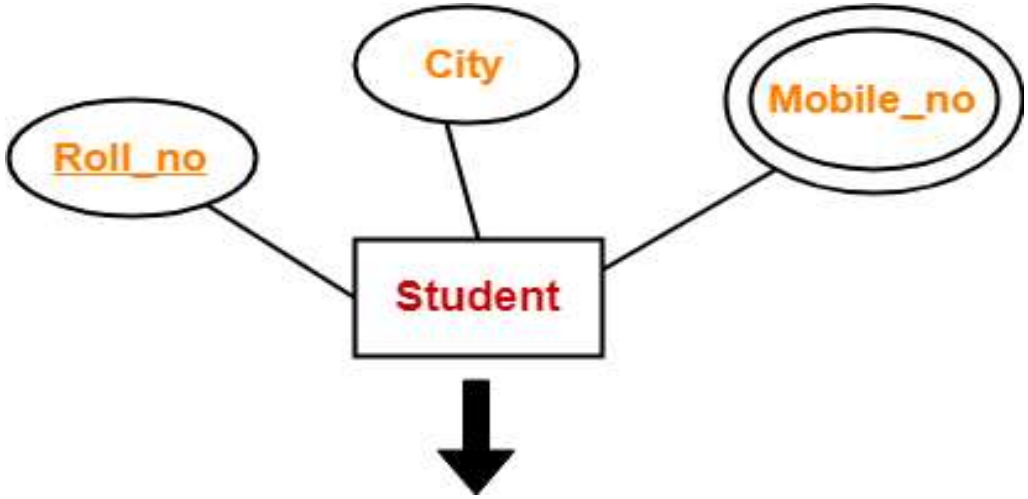
Schema : Student (Roll_no , First_name , Last_name , House_no , Street , City)

Rules for Conversion

Rule-03: For Strong Entity Set With Multi Valued Attributes-

- *A strong entity set with any number of multi valued attributes will require two tables in relational model.*
- *One table will contain all the simple attributes with the primary key.*
- *Other table will contain the primary key and all the multi valued attributes.*

Explanation with Example



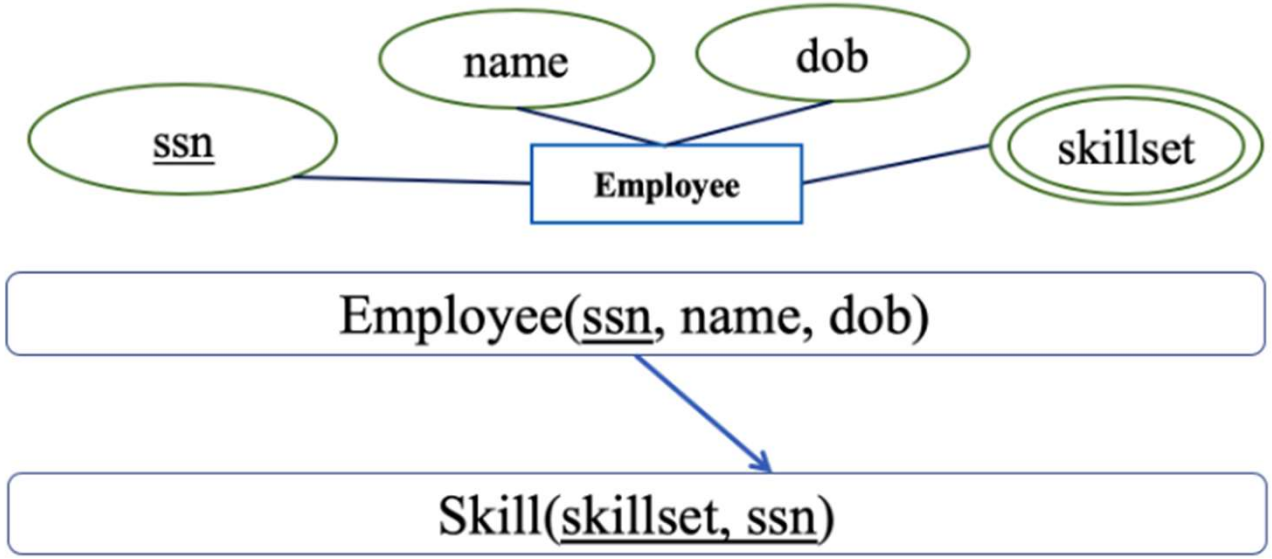
Roll_no	City
PK	

Roll_no	Mobile_no
PK,FK	PK

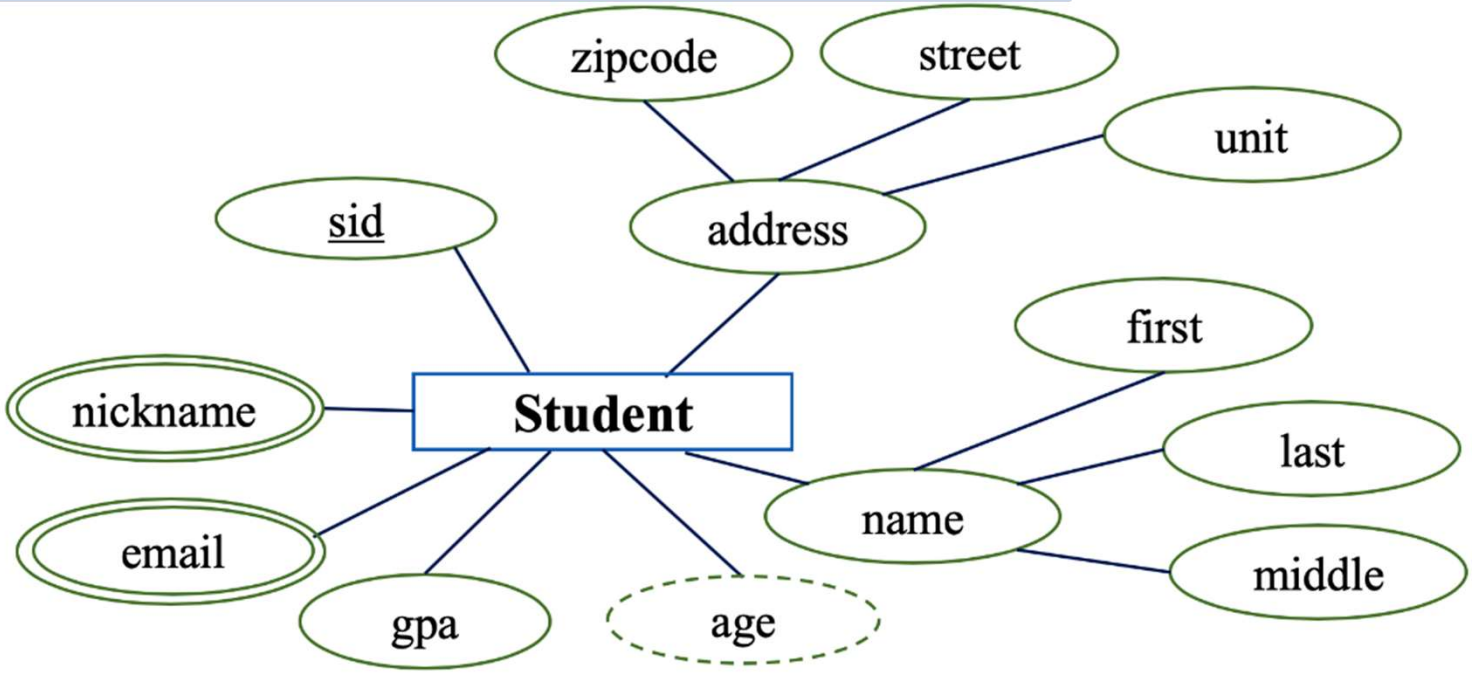
Roll_no	City

Roll_no	Mobile_no

Explanation with Example



Explanation with Example



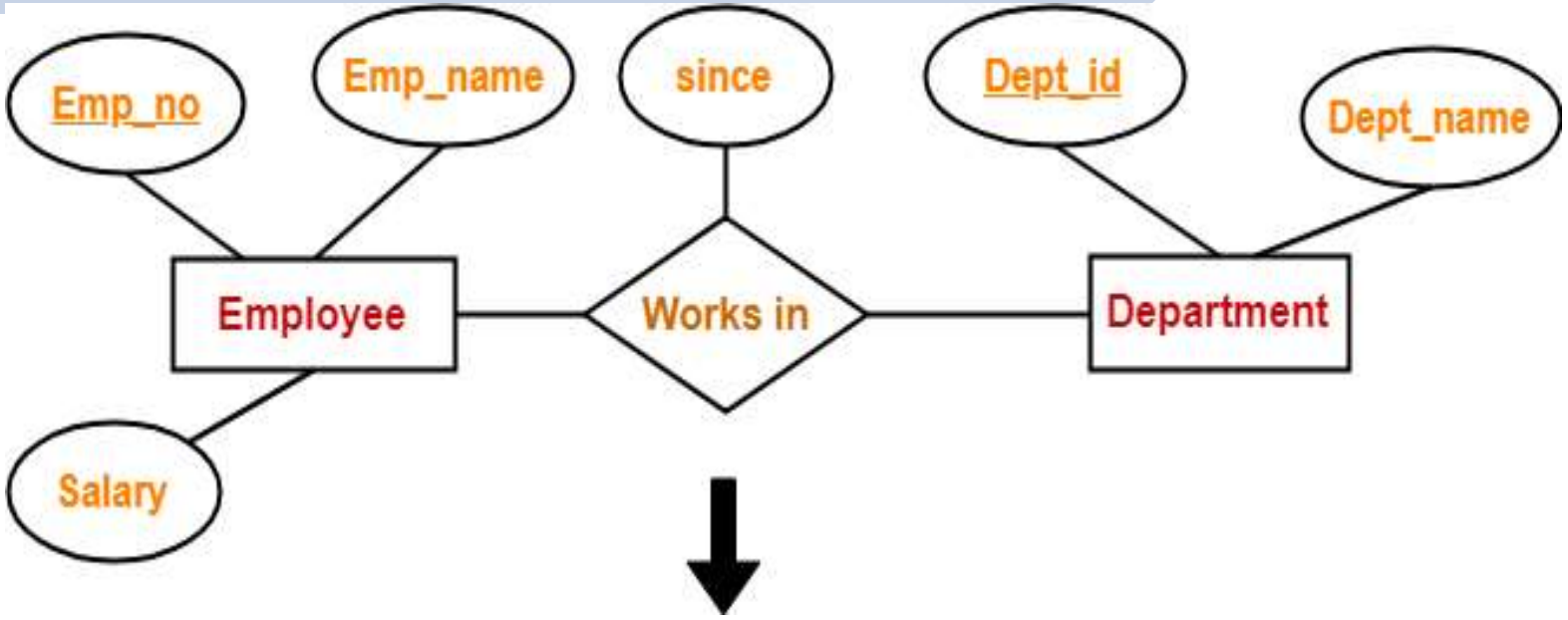
Student(<u>sid</u>, zipcodeAddr, streetAddr, unitAddr, firstName, middleName, lastName, gpa, dob)
Nickname(<u>sid</u>, <u>nickname</u>)
Email(<u>sid</u>, <u>email</u>)

Translating Relationship into Table

Rule-04: Translating Relationship Set into a Table-

- *A relationship set will require one table in the relational model*
- *Attributes of the table are-*
  - *-Primary key attributes of the participating entity sets*
  - *-Its own descriptive attributes if any*
- *Set of non-descriptive attributes will be the primary key*

Explanation with Example



Emp_no	Dept_id	since

Schema : Works in (Emp_no , Dept_id , since)

If we consider the overall ER diagram, three tables will be required in relational model-

-One table for the entity set "Employee"

-One table for the entity set "Department"

-One table for the relationship set "Works in"

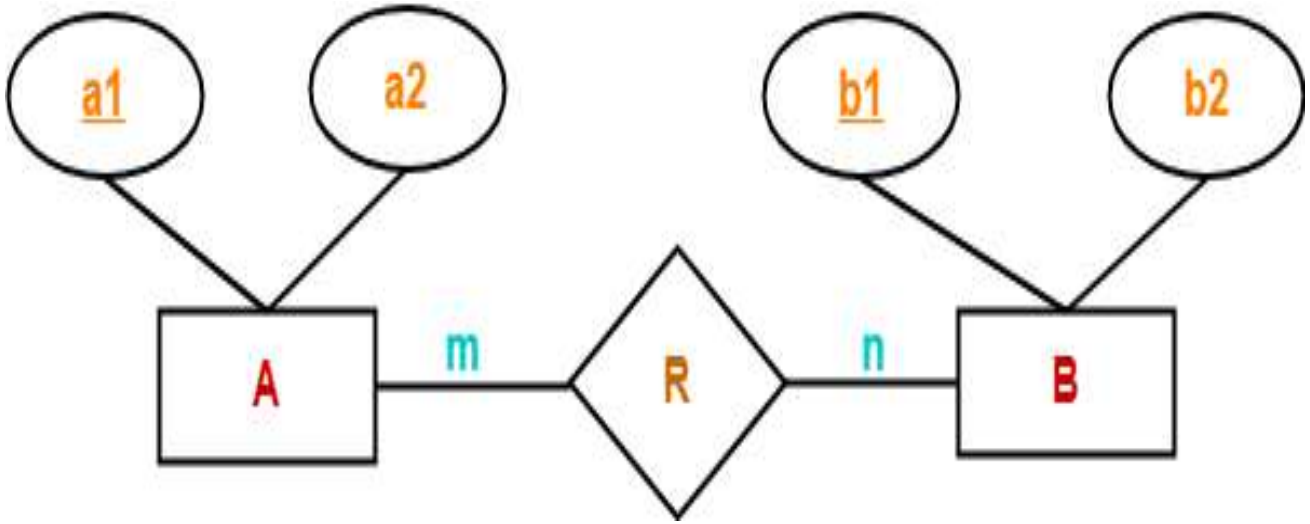


Binary Relationship with Cardinality Ratios

Four possible cases:

- ✓**Case-01:** Binary relationship with cardinality ratio m:n
- ✓**Case-02:** Binary relationship with cardinality ratio 1:n
- ✓**Case-03:** Binary relationship with cardinality ratio m:1
- ✓**Case-04:** Binary relationship with cardinality ratio 1:1

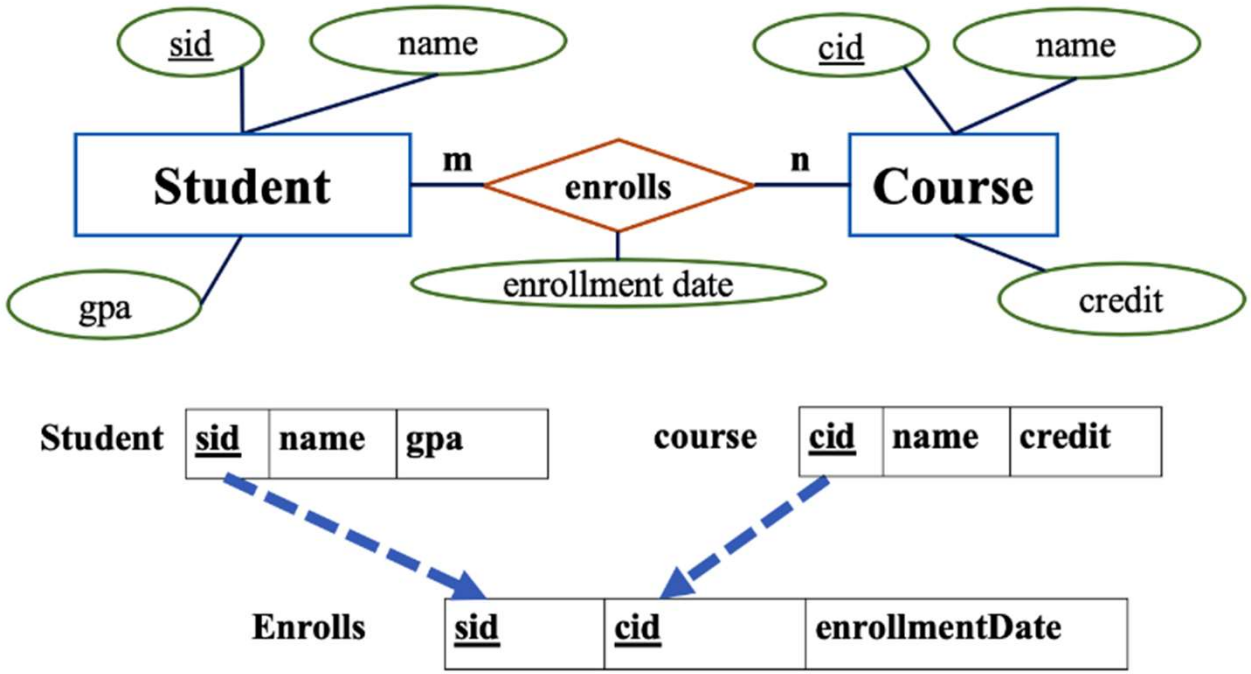
Binary relationship with cardinality ratio m:n



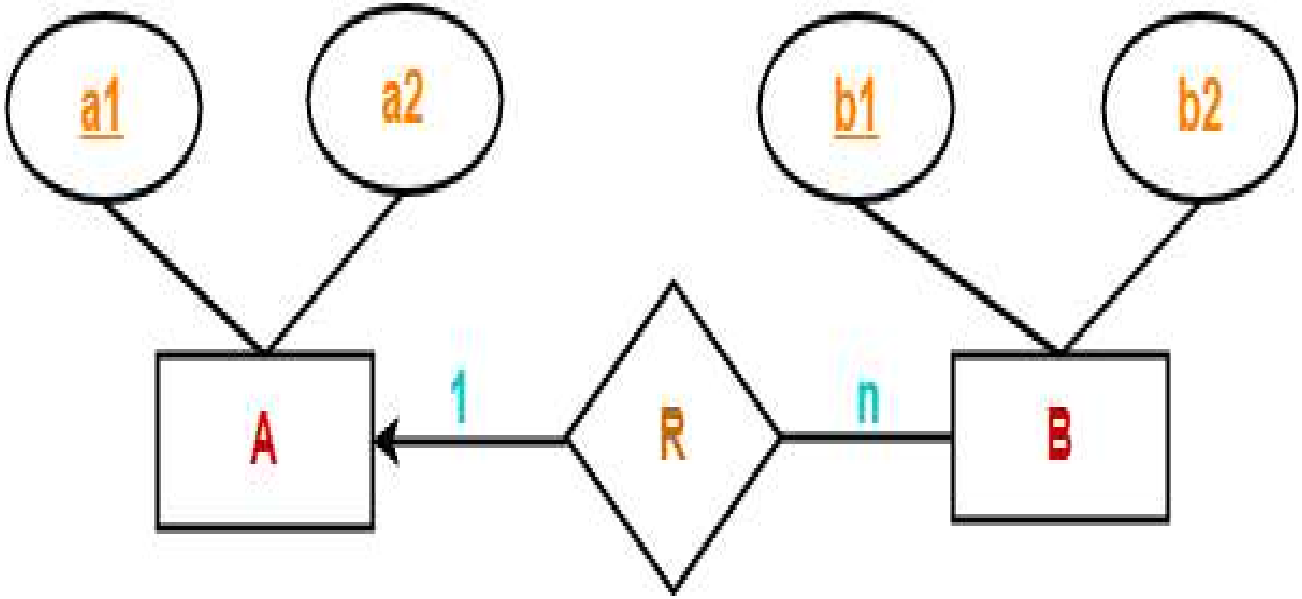
Here, three tables will be required-

A (a1 , a2)
R (a1 , b1)
B (b1 , b2)

Binary relationship with cardinality ratio m:n



Binary relationship with cardinality ratio 1:n



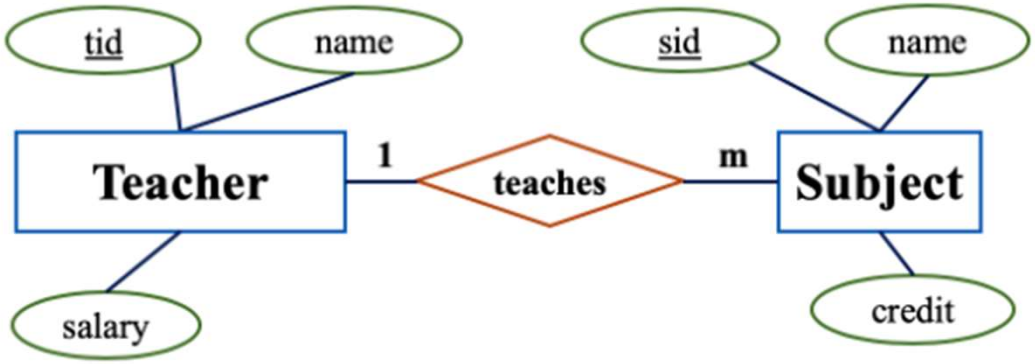
Here, two tables will be required-

A (a1 , a2)

BR (a1 , b1 , b2)

NOTE- Here, combined table will be drawn for the entity set B and relationship set R

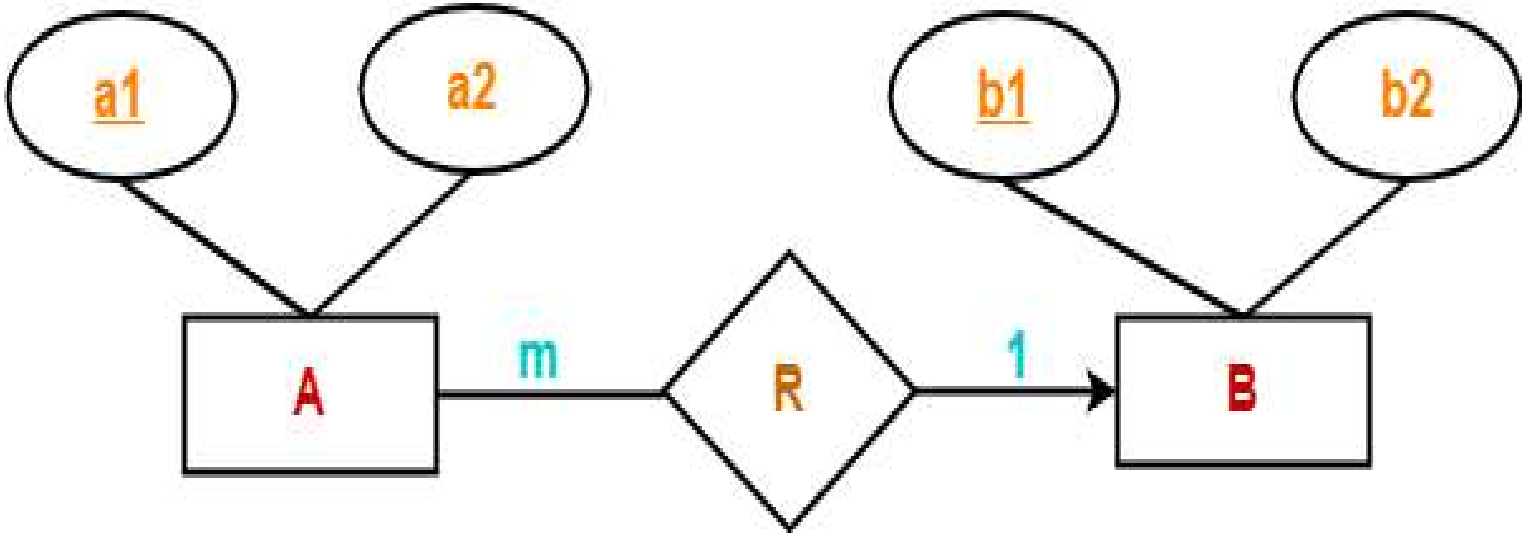
Binary relationship with cardinality ratio 1:n



Teacher	tid	name	salary

Subject	sid	name	credit	teacherId

Binary relationship with cardinality ratio m:1



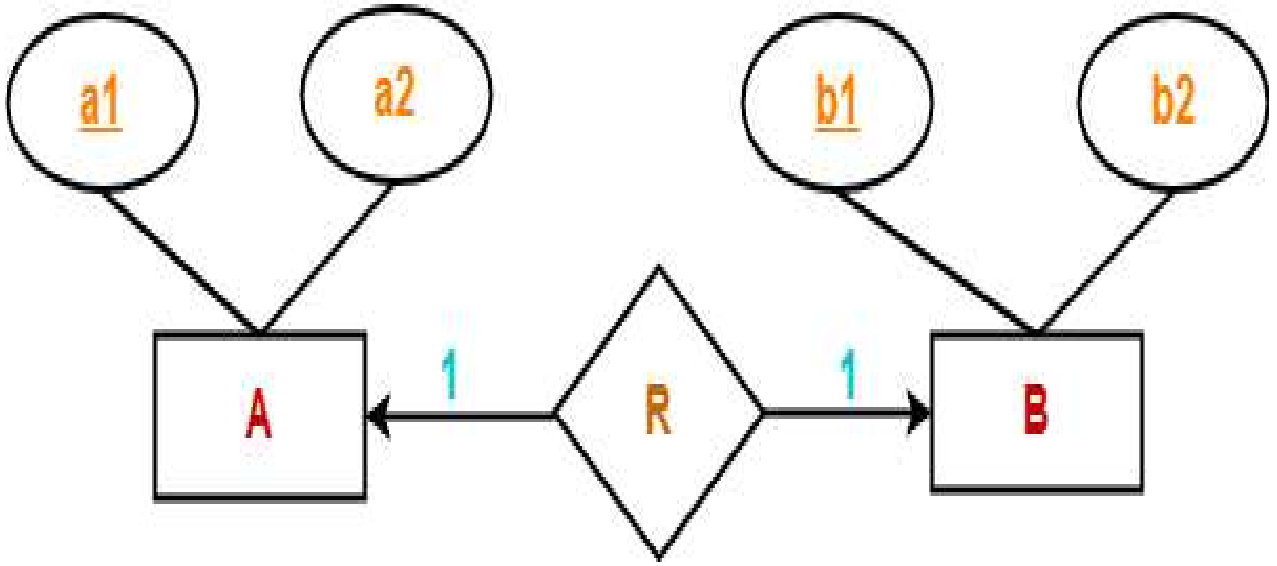
Here, two tables will be required-

AR (a1 , a2 , b1)

B (b1 , b2)

NOTE- Here, combined table will be drawn for the entity set A and relationship set R

Binary relationship with cardinality ratio 1:1



Here, two tables will be required. Either combine 'R' with 'A' or 'B'

Way-01:

AR (a1 , a2 , b1)

B (b1 , b2)

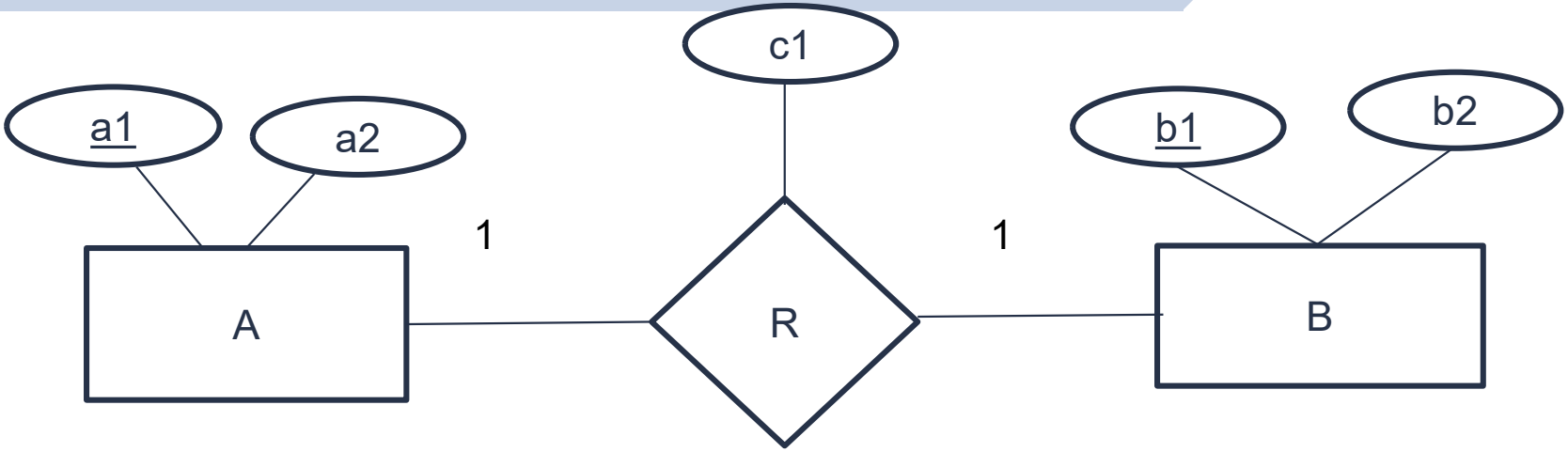
Way-02:

A (a1 , a2)

BR (a1 , b1 , b2)

Tabular separation of relationship

1:1 (CASE I)



Both side partial

A

a1	a2	b1	c1

B

b1	b2

OR

A

a1	a2

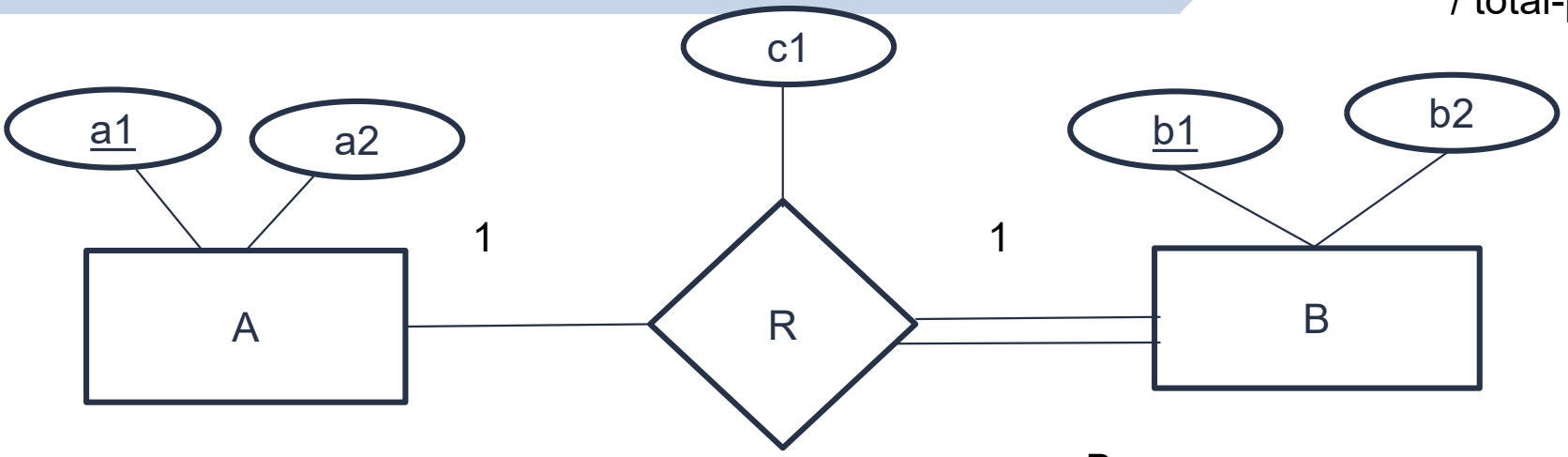
B

b1	b2	c1	a1

Move b1 to left side entity set A OR Move a1 to right side entity set B

Tabular separation of relationship

1:1 (CASE II) with partial-total / total-partial



A

a1	a2

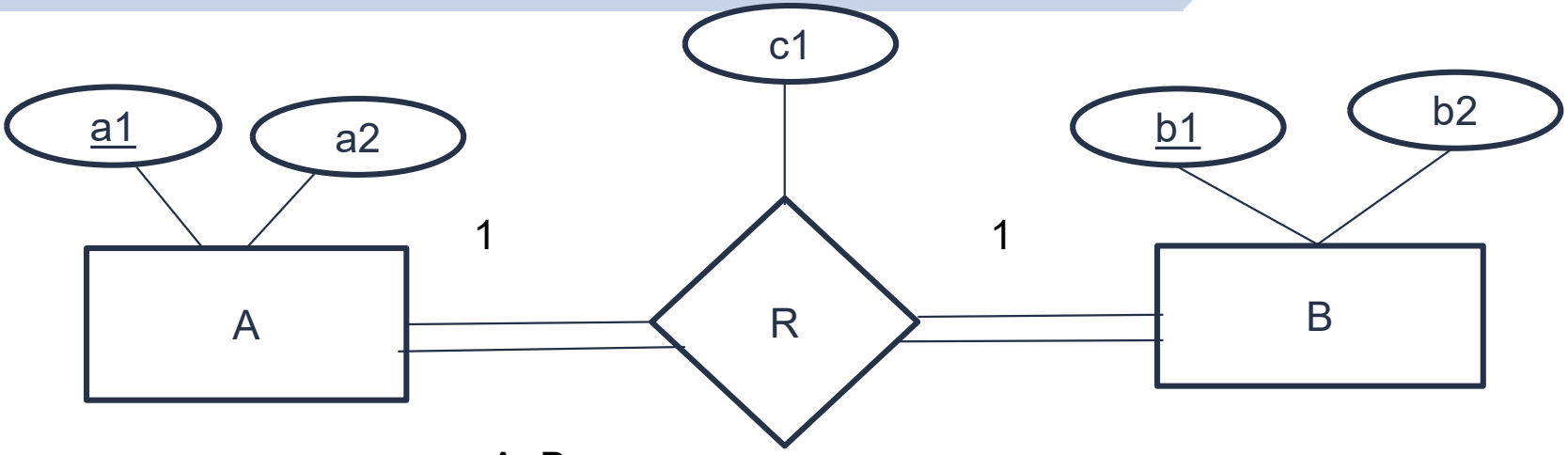
B

b1	b2	c1	a1

In all the cases move the attributes of relation R to the total side entity set

Tabular separation of relationship

1:1 (CASE III) total = total



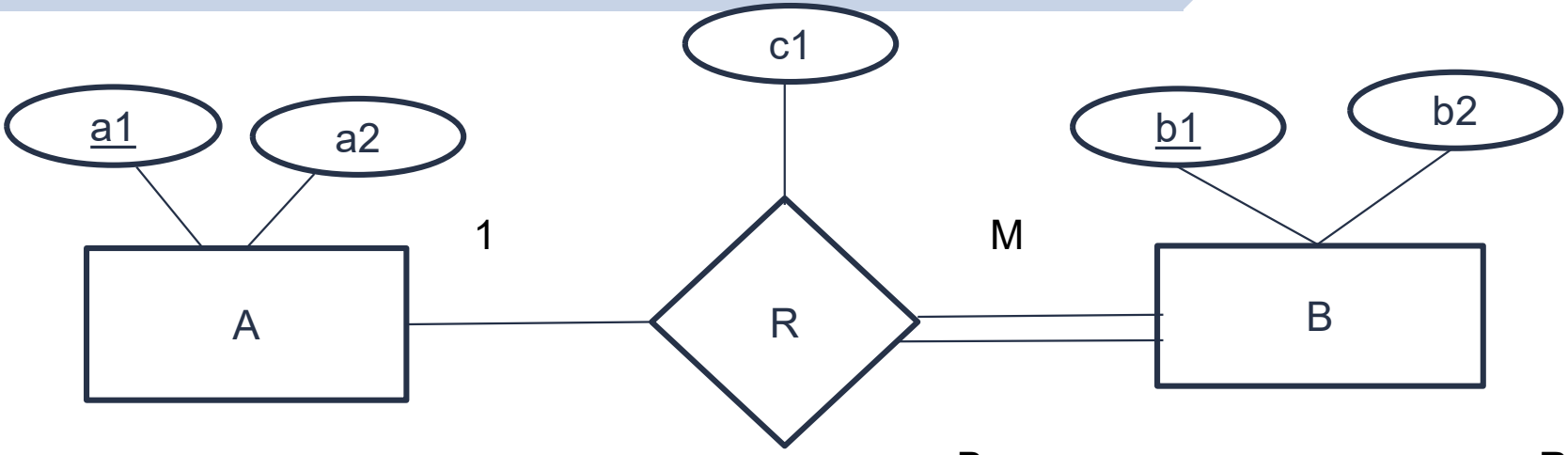
A_B

a1	a2	b1	b2	c1

Merge all to a single table and take either a1 or b1 as key

Tabular separation of relationship

1:M with all participations



A

a1	a2

B

b1	b2	a1	c1

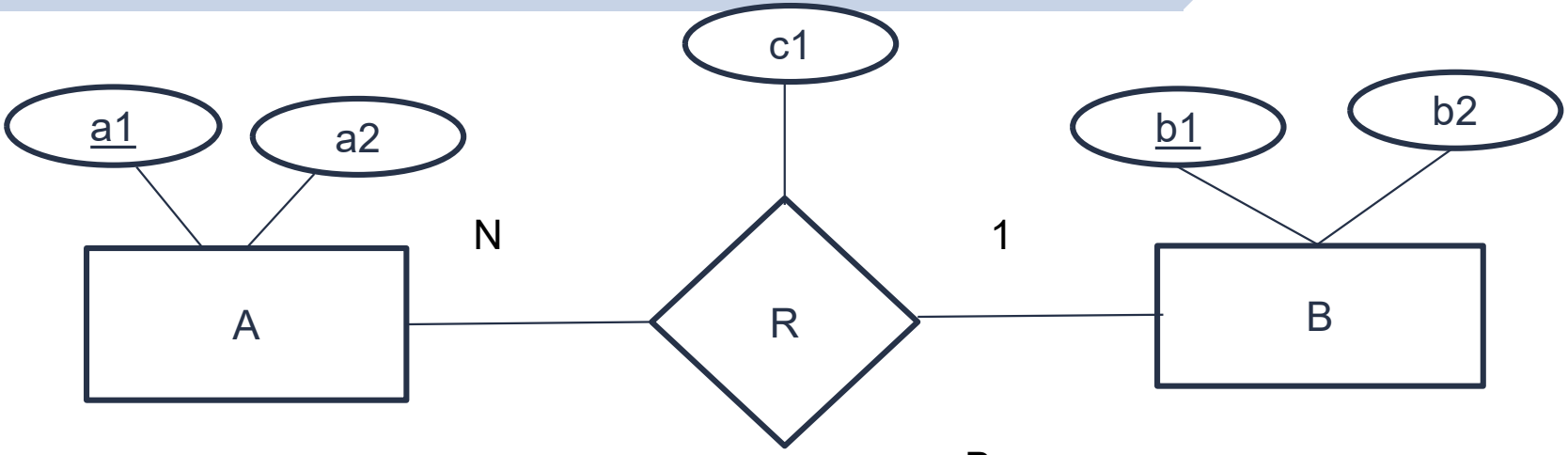
R

a1	b1	c1

Replace the attribute of relationship R to many entity set irrespective of participation

Tabular separation of relationship

N:1 with all participations



A

a1	a2	b1	c1

B

b1	b2

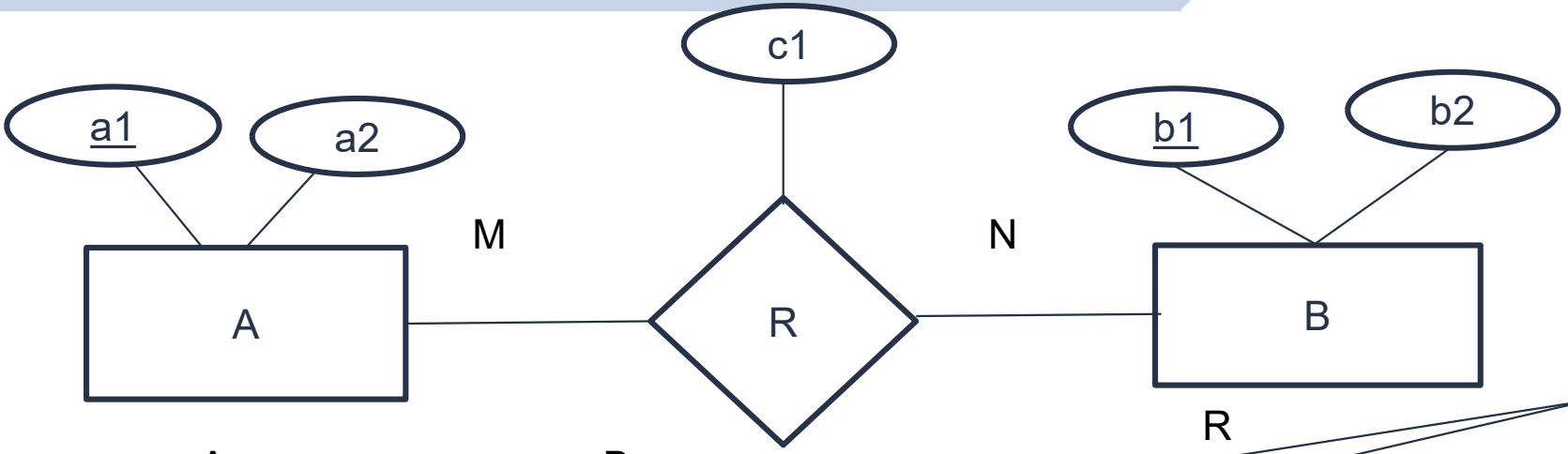
R

a1	b1	c1

Replace the attribute of relationship R to many entity set irrespective of participation

Tabular separation of relationship

M:N with all participations



A

a1	a2

B

b1	b2

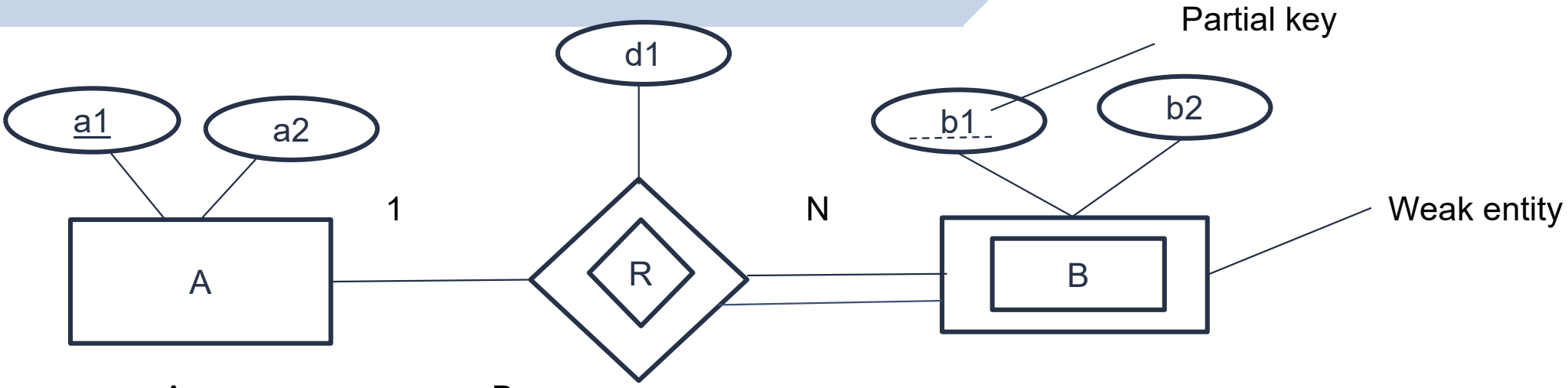
R

a1	b1	c1

Composite primary key

Create a separate table for relationship R where the primary key will be composite

Tabular separation of relationship



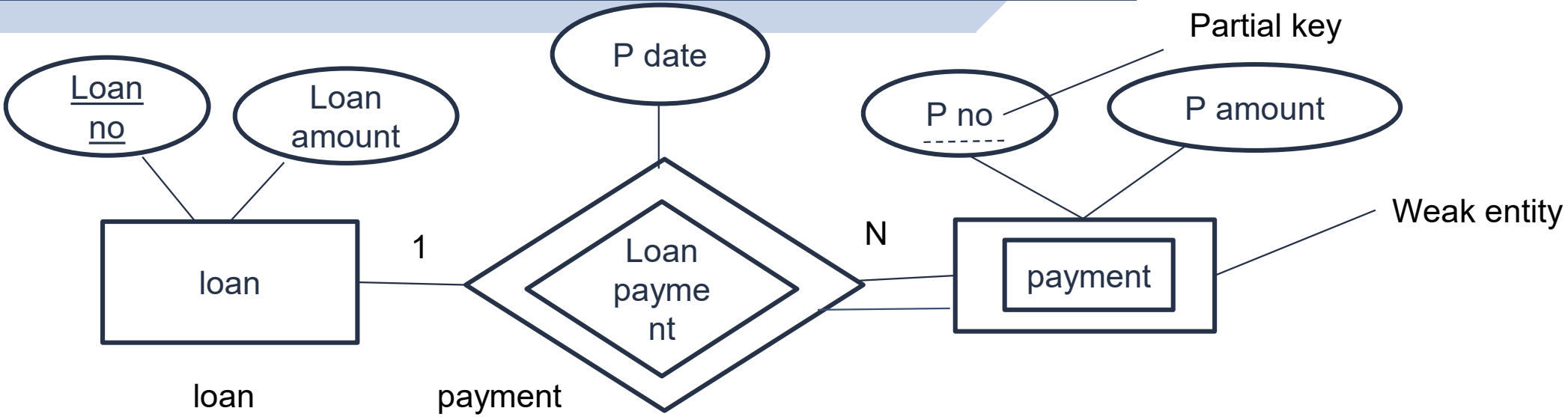
A

a1	a2

B

a1	b1	b2	d1

Example Tabular separation of relationship



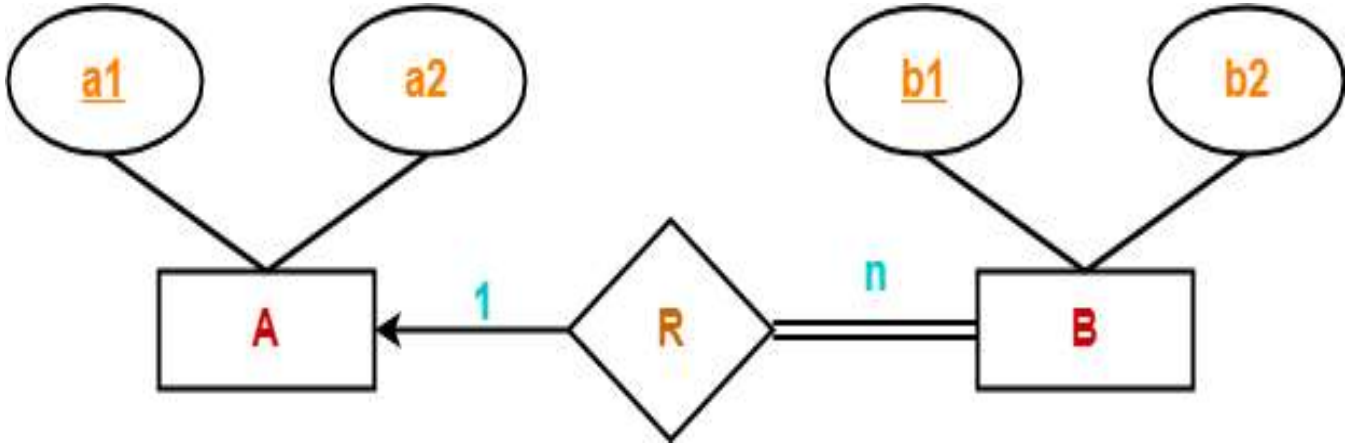
loan

Loan no	Loan amount

payment

Loan no	P no	P amount	P date

Binary relationship with cardinality constraint and total participation from one side



Because cardinality ratio = 1 : n , so we will combine the entity set B and relationship set R.

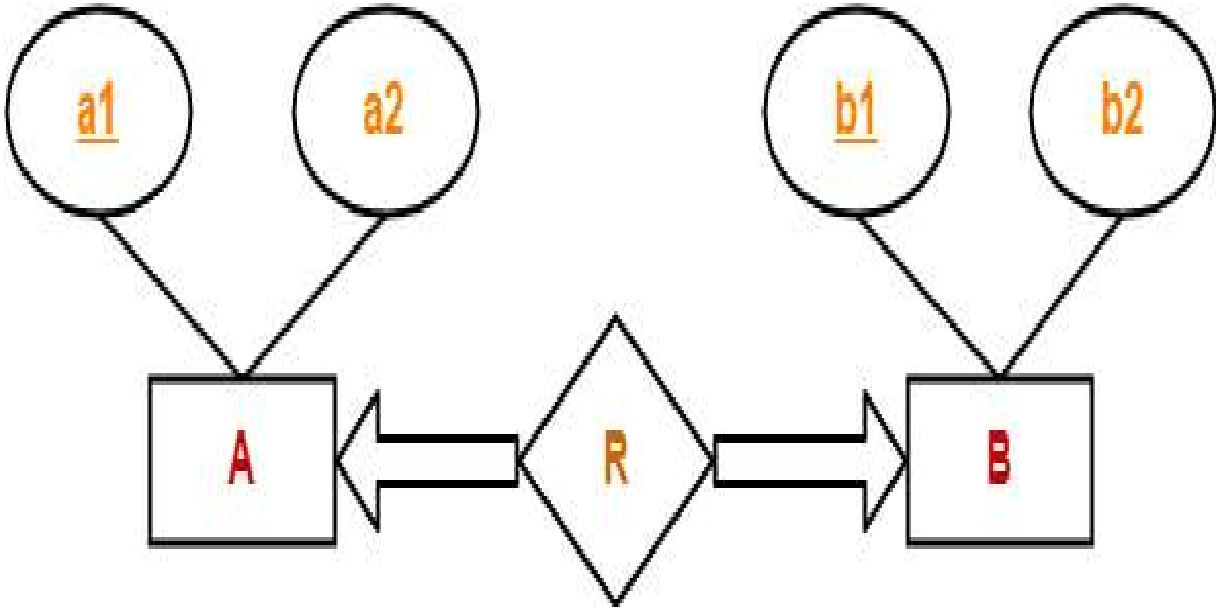
Then, two tables will be required-

A (a1 , a2)

BR (a1 , b1 , b2)

Because of total participation, foreign key a1 has acquired NOT NULL constraint, so it can't be null now.

Binary relationship with cardinality constraint and total participation from both sides

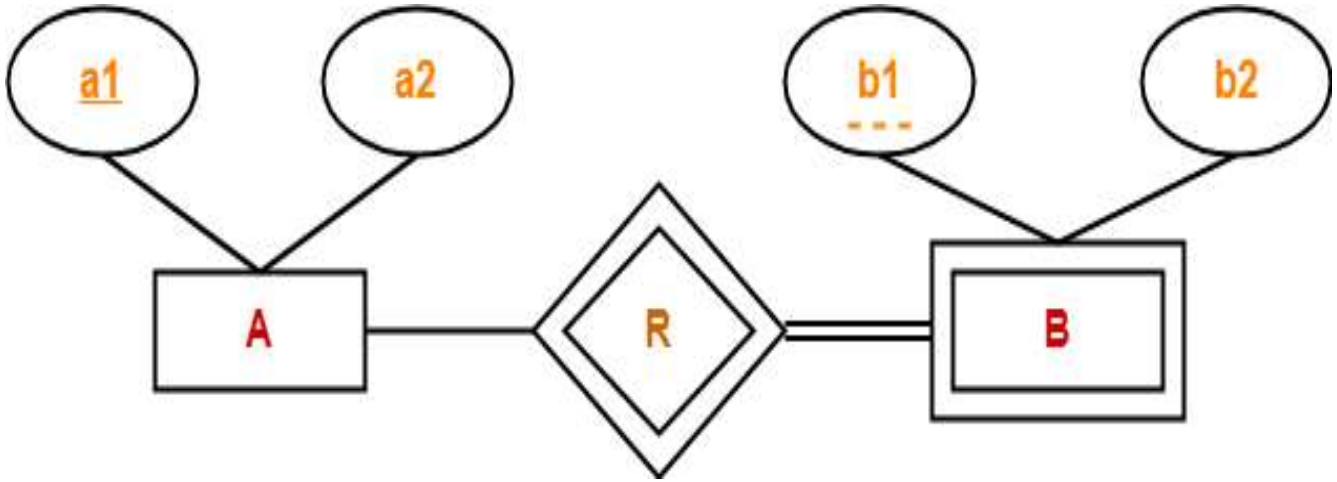


Here, Only one table is required.
ARB (a1 , a2 , b1 , b2)

If there is a key constraint from both the sides of an entity set with total participation, then that binary relationship is represented using only single table

Binary relationship with Weak Entity Set

Weak entity set always appears in association with identifying relationship with total participation constraint.

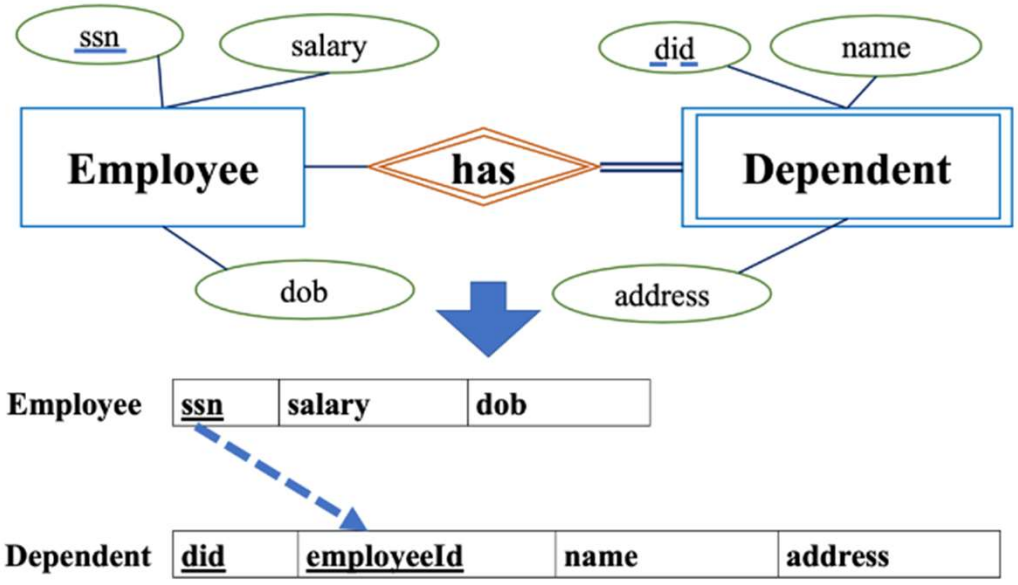


Here, two tables will be required-

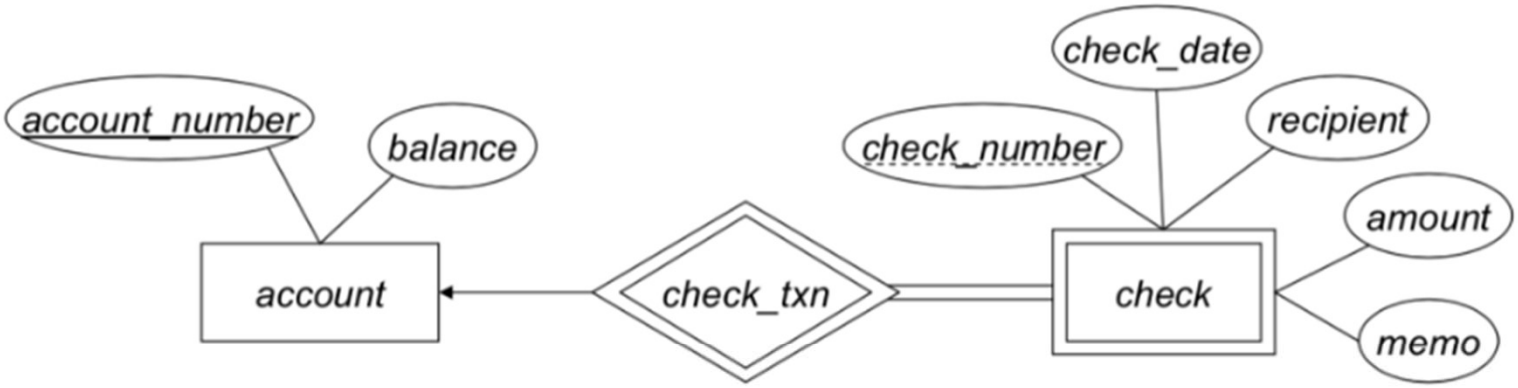
A (a1 , a2)

BR (a1 , b1 , b2)

Binary relationship with Weak Entity Set



Binary relationship with Weak Entity Set



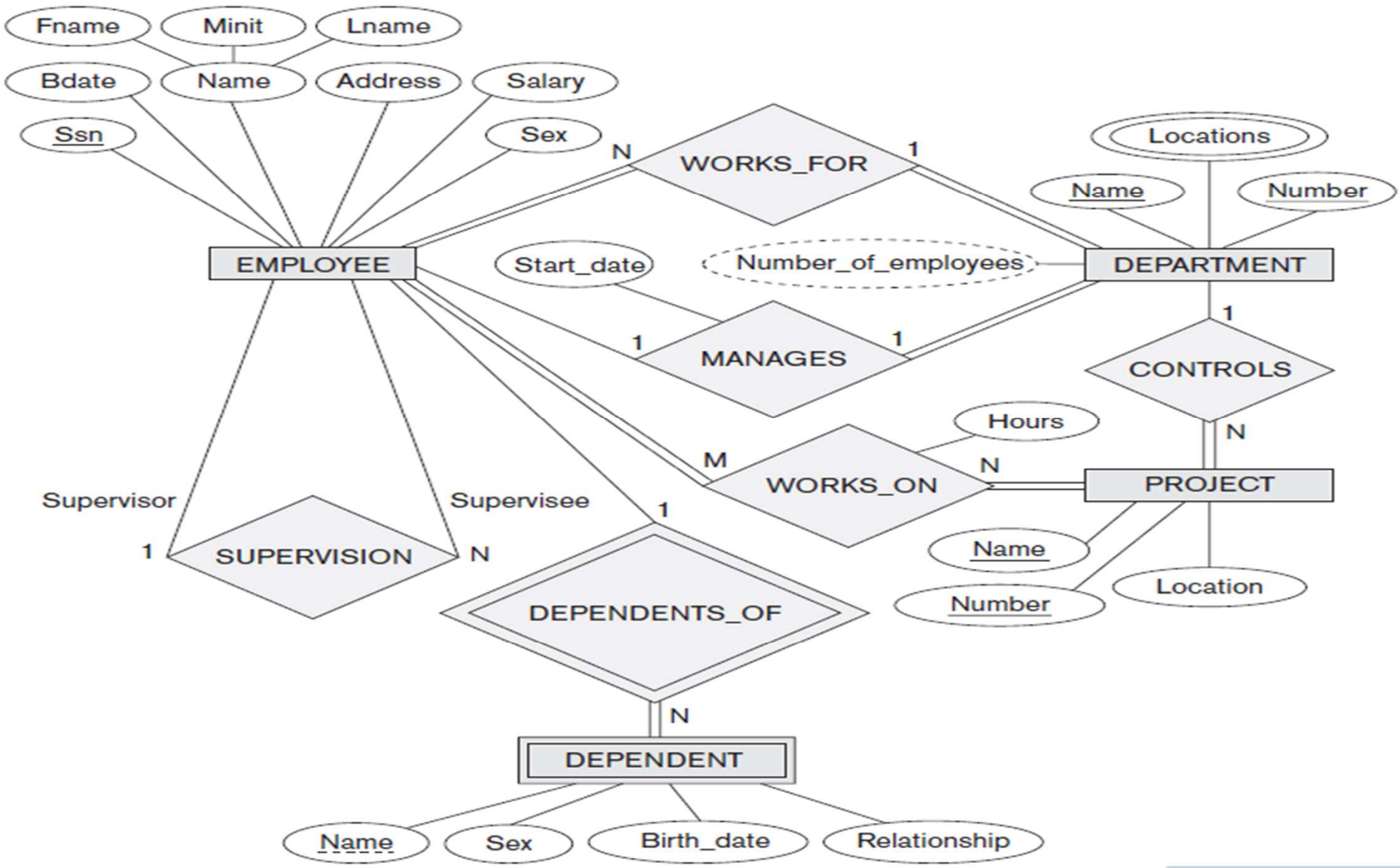
account(*account_number*, *balance*)

check schema:

Discriminator is *check_number*

Primary key for *check* is: (*account_number*, *check_number*)

check(*account_number*, *check_number*, *check_date*, *recipient*, *amount*, *memo*)


Fname
Minit
Lname
Name
Bdate
Address
Salary
Ssn
Sex
N
WORKS_FOR
1
Locations
Name
Number
EMPLOYEE
Start_date
Number_of_employees
DEPARTMENT
1
1
MANAGES
1
CONTROLS
Hours
N
M
WORKS_ON
N
PROJECT
Supervisor
Supervisee
1
1
SUPERVISION
N
Name
Location
Number
DEPENDENTS_OF
N
DEPENDENT
Name
Sex
Birth_date
Relationship

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date

DEPT_LOCATIONS

Dnumber	Dlocation

PROJECT

Pname	Pnumber	Plocation	Dnum

WORKS_ON

Essn	Pno	Hours

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship

Thank You

Any questions?

